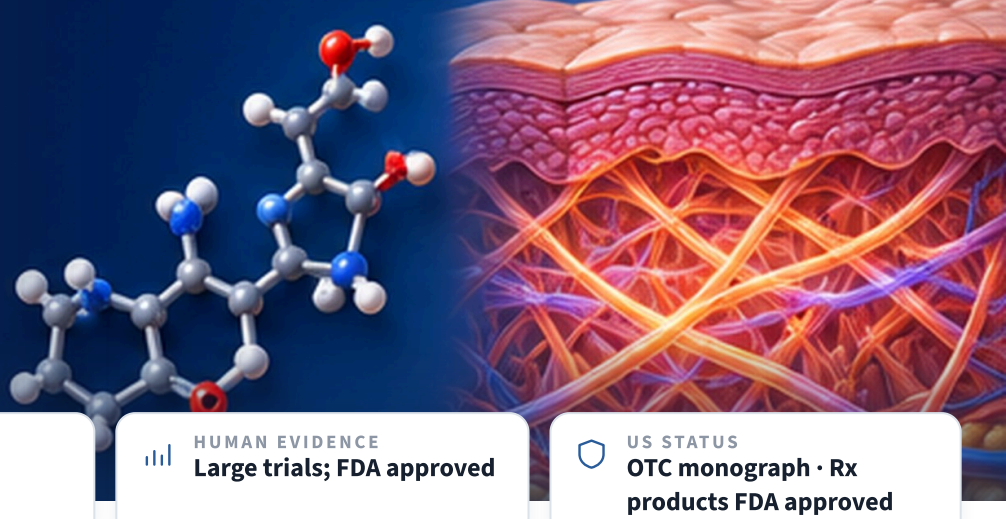
SKIN 11

Benzoyl peroxide

Oxidizing agent applied to the skin; converted to benzoic acid in the skin



CLASS
Topical drug (oxidizing agent)

STUDIED FOR
Acne, rosacea

HUMAN EVIDENCE
Large trials; FDA approved

US STATUS
OTC monograph · Rx products FDA approved

STUDIED FOR

Acne	Rosacea
HUMAN	HUMAN

Most trials were short. Most acne trials ran up to 12 weeks, and many had a high or unclear risk of bias (Cochrane, 2020).

HOW IT WORKS IN THE BODY

Benzoyl peroxide → Oxidizing agent → Kills bacteria → Loosens skin layer

- **Oxidizing agent** with bactericidal and keratolytic effects (label)
- **In the skin** it is converted to benzoic acid, which leaves the body in urine
- **In rosacea,** the precise mechanism of action is unknown (label)

EVIDENCE & WHAT'S KNOWN

EVIDENCE

FDA APPROVED
LEVEL 5 OF 5

- Cochrane 2020: 120 trials, most up to 12 weeks
- Epsolay: 2 randomized trials, 733 adults
- OTC acne monograph at 2.5% to 10%

NOT YET ESTABLISHED

- Amount absorbed into the body
- Meaning of animal tumor findings
- Effects on fertility

QUICK PRACTICAL INFO

HALF-LIFE
Not established (body exposure not assessed)

ROUTES STUDIED
On the skin (creams, gels, washes)

PHARMACOKINETICS
Converted to benzoic acid in skin; eliminated in urine

SOLD AS
OTC 2.5% to 10%; Rx creams and combination gels

MONITORED IN TRIALS
Skin irritation, allergic reactions

SPORT (WADA)
Not on the 2026 list

REGULATORY

- FDA — OTC acne monograph, 2.5% to 10% (21 CFR 333)
- FDA — Rx Epsolay (2022) and combinations approved
- FDA — Benzene found in 6 of 95 products tested (2025)
- WADA — Not listed

REFERENCES

1. FDA, Epsolay (benzoyl peroxide) cream 5% label, Aug 2025
2. Cochrane Database Syst Rev, topical benzoyl peroxide for acne, Mar 2020
3. WADA, 2026 Prohibited List

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